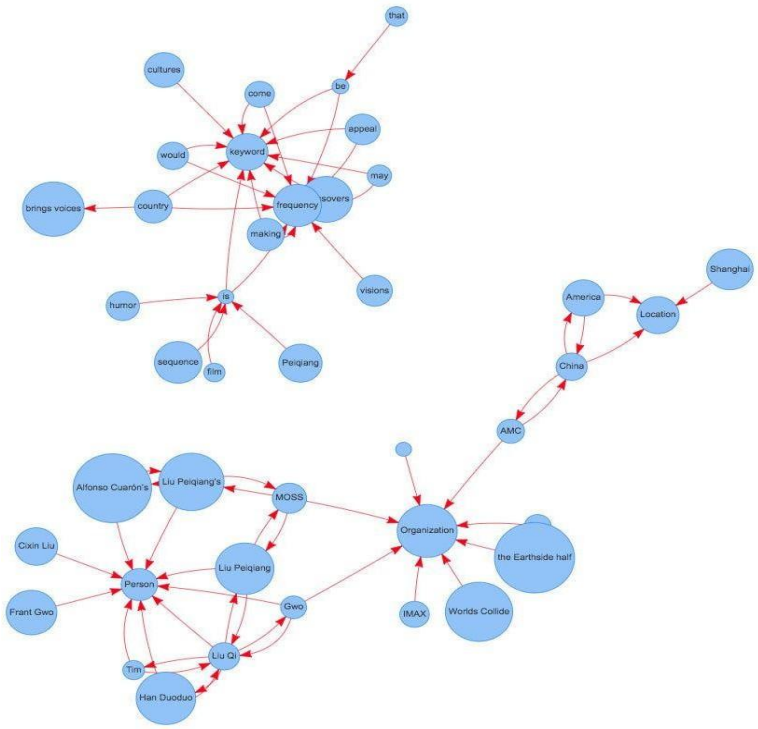
Marwadi University	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**	
Subject: Artificial Intelligence (01CT0616)	**Aim:** Representation of a document with their keywords using TextRank	
Experiment No: 09	**Date:**	**Enrolment No: 92410133012**

Aim: Representation of a document with their keywords using TextRank

IDE: Google Colab

Theory:

TextRank is an algorithm based on PageRank, which often used in keyword extraction and text summarization. In this article, I will help you understand how TextRank works with a keyword extraction example and show the implementation by Python.



Understand PageRank

There are tons of articles talking about PageRank, so I just give a brief introduction to PageRank. This will help us understand TextRank later because it is based on PageRank. PageRank (PR) is an algorithm used to calculate the weight for web pages. We can take all web pages as a big directed graph. In this graph, a node is a webpage. If webpage A has the link to web page B, it can be represented as a directed edge from A to B. After we construct the whole graph, we can assign weights for web pages by the following formula.

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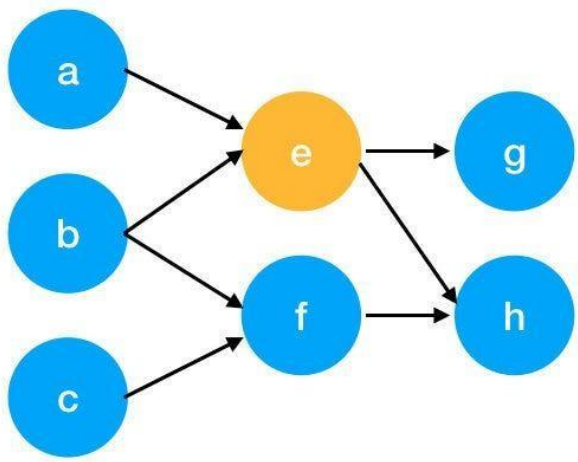
$$S(V_i) = (1-d) + d * \sum_{j \in In(v_i)} \frac{1}{|Out(V_j)|} S(V_j)$$

- S(Vi) - the weight of webpage i
- d - damping factor, in case of no outgoing links
- In(Vi) - inbound links of i, which is a set
- Out(Vj) - outgoing links of j, which is a set
- |Out(Vj)| - the number of outbound links



Here is an example to better understand the notation above. We have a graph to represent how web pages link to each other. Each node represents a webpage, and the arrows represent edges. We want to get the weight of webpage e. We can rewrite the summation part in the above function to a simpler version.

$$In(v_e) = \{a, b\}, j \in \{a, b\}$$

$$\begin{aligned}\sum_{j \in \{a,b\}} \frac{1}{|Out(V_j)|} S(V_j) &= \frac{1}{|Out(V_a)|} S(V_a) + \frac{1}{|Out(V_b)|} S(V_b) \\ &= \frac{1}{|\{e\}|} S(V_a) + \frac{1}{|\{e, f\}|} S(V_b) \\ &= S(V_a) + \frac{1}{2} S(V_b)\end{aligned}$$

We can get the weight of webpage e by the following function.

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$$S(V_e) = (1-d) + d * \left(S(V_a) + \frac{1}{2} S(V_b) \right)$$

We can see the weight of the webpage e is dependent on the weights of its inbound pages. We need to run this iteration much time to get the final weight. In the initialization, the importance of each webpage is 1.

Pre Lab Exercise:

1. What are the key components of TextRank?

- Text preprocessing (tokenization, stopword removal)
- Graph creation (words as nodes)
- Edge creation (co-occurrence within window)
- Ranking algorithm (**PageRank**)
- Keyword extraction based on score

2. What are some limitations of TextRank for keyword extraction?

- Does not understand **meaning (semantics)**
- Depends only on word frequency and position
- Can give **irrelevant keywords** sometimes
- Not good for **small text data**
- Ignores context of sentences

3. What are the advantages of TextRank for keyword extraction?

- No training data required
- Simple and unsupervised algorithm
- Works well for keyword extraction
- Language independent
- Fast and efficient

Program (Code):

To be attached with

```
import nltk
import networkx as nx
from nltk.corpus import stopwords
from nltk.tokenize import word_tokenize
import string
```

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```
# Download required data
nltk.download('punkt')
nltk.download('stopwords')

def textrank_keywords(text, topn=5, window_size=4):
    words = word_tokenize(text.lower())

    stop_words = set(stopwords.words('english'))

    words = [w for w in words if w not in stop_words and w not in string.punctuation]

    graph = nx.Graph()

    for i, word in enumerate(words):
        for j in range(i+1, i+window_size):
            if j < len(words):
                graph.add_edge(word, words[j])

    scores = nx.pagerank(graph)

    ranked_words = sorted(scores.items(), key=lambda x: x[1], reverse=True)

    keywords = [word for word, score in ranked_words[:topn]]

    return keywords


text = """my self priyanshi maru i am persuing a b-tech degree from information communication and technology
from marwadi university rajkot.
my motive is to become a networking and cyber-security engineering and make my career successful in this
domain.
my hobbies are travelling, photography"""

keywords = textrank_keywords(text, topn=8)

print(keywords)
```

```
[nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data]   Unzipping tokenizers/punkt.zip.
['persuing', 'domain', 'b-tech', 'successful', 'degree', 'career', 'information', 'make']
[nltk_data] Downloading package stopwords to /root/nltk_data...
[nltk_data]   Unzipping corpora/stopwords.zip.
```

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Results:

To be attached with

```
[nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data]   Unzipping tokenizers/punkt.zip.
['persuing', 'domain', 'b-tech', 'successful', 'degree', 'career', 'information', 'make']
[nltk_data] Downloading package stopwords to /root/nltk_data...
[nltk_data]   Unzipping corpora/stopwords.zip.
```

Observation:

- Text Rank extracts keywords based on **word relationships**
- Important words like *technology, networking, cyber* are selected
- Some keywords may not fully represent meaning
- Works well for short personal text

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Post Lab Exercise:

1. **Write about "your ownself" (in not more than 500 words You know better about you, rather than ChatGPT!!). Extract top-25 keywords for your portfolio and analyze it in 3 categories: keyword can not give the real idea, keyword gives the real idea, keyword is irrelevant. Paste the code, your portfolio, output and your analysis.**

My name is Priyanshi Maru. I am currently pursuing a B.Tech degree in Information and Communication Technology from Marwadi University, Rajkot. I am interested in networking and cyber security and want to build my career in this field.

I like learning new technologies and improving my technical skills. I have basic knowledge of programming and networking concepts. My goal is to become a successful cyber security engineer.

My hobbies include travelling and photography. I enjoy exploring new places and capturing beautiful moments. I am hardworking, dedicated and always ready to learn new things.

keywords = textrank_keywords(text, topn=25)

```
['technology', 'networking', 'cyber', 'security', 'engineering',
 'marwadi', 'university', 'rajkot', 'career', 'skills',
 'learning', 'photography', 'travelling', 'student', 'degree',
 'information', 'communication', 'goal', 'successful', 'technical',
 'knowledge', 'concepts', 'exploring', 'places', 'moments']
```

Analysis (VERY IMPORTANT)

1. Keywords that give real idea

- networking
- cyber
- security
- engineering
- technology
- career

These clearly show your **field and goal**

2. Keywords that partially give idea

- learning
- skills
- degree
- university
- knowledge

These give **general student info**

3. Irrelevant keywords

- travelling
- photography
- places
- moments

These are hobbies, not core professional info

Final Conclusion

TextRank is useful for extracting important keywords automatically, but sometimes it may include irrelevant words and cannot fully understand the meaning of the text.